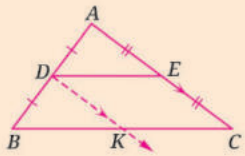
■ Discuss

How can you prove that the length of the midsegment of a triangle is equal to half the length of its opposite side?

■ Hint

Draw $\overrightarrow{DK}$ parallel to $\overline{AC}$ and intersecting $\overline{BC}$ at K, then prove that $DKCE$ is a parallelogram.

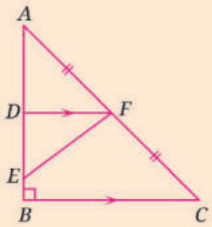


■ Self-Evaluation (2)

In the following figure:

$AB = BC = 16\ cm$

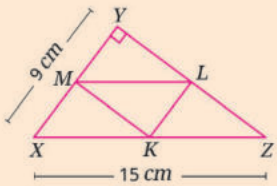
$EB = 2\ cm$



Find the length of $\overline{EF}$.

■ Self-Evaluation (3)

In the following figure:



The sides of the triangle KLM are the midsegments of the triangle XYZ, which is right angled at Y. Calculate the perimeter of the triangle KLM.

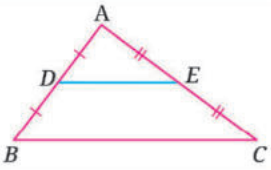
Theorem 3-3 (Theorem of the midsegment of a triangle)

The midsegment of a triangle is parallel to the third side and its length is equal to half the length of that side.

If $\overline{DE}$ is a midsegment of $\Delta\ ABC$,

then : (1) $\overline{DE} \parallel \overline{BC}$

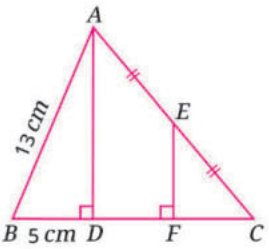
(2) $DE = \frac{1}{2} BC$



Example 2

In the given figure:

Find with proof the length of $\overline{EF}$.



$\because \Delta\ ABD$ is right-angled at D

$\therefore (AD)^2 = (AB)^2 - (BD)^2 = (13)^2 - (5)^2 = 144$

$\therefore AD = \sqrt{144} = 12$

$\because m(\angle ADB) = m(\angle EFB) = 90°$ (and they are corresponding angles)

$\therefore \overline{AD} \parallel \overline{EF}$

In $\Delta\ ADC$: $\because E$ is the midpoint of $\overline{AC}$, $\overline{AD} \parallel \overline{EF}$

$\therefore F$ is the midpoint of $\overline{DC}$

$\therefore EF = \frac{1}{2} AD$ $\quad\quad$ $\therefore EF = \frac{1}{2} \times 12 = 6\ cm.$

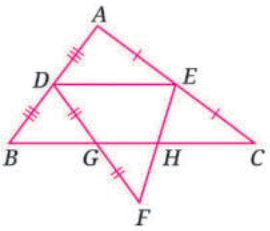
Example 3

In the given figure:

If D is the midpoint of $\overline{AB}$, E is the midpoint of $\overline{AC}$,

G is the midpoint of $\overline{DF}$,

prove that: $BC = 4\ GH$



In $\Delta\ ABC$:

$\because D$ is the midpoint of $\overline{AB}$, E is the midpoint of $\overline{AC}$

$\therefore \overline{DE} \parallel \overline{BC}$, $DE = \frac{1}{2} BC$

In $\Delta\ FDE$:

$\because G$ is the midpoint of $\overline{DF}$, $\overline{DE} \parallel \overline{GH}$ $\quad\quad$ $\therefore H$ is the midpoint of $\overline{EF}$

$\therefore GH = \frac{1}{2} DE$ $\quad\quad$ $\therefore GH = \frac{1}{2}(\frac{1}{2} BC) = \frac{1}{4} BC$ $\quad\quad$ $\therefore BC = 4\ GH$